

Getting your keyboard capture-ready for Aether

A step-by-step guide for board submitters -
do it yourself, or have an AI do it for you.

You submitted your keyboard to Aether (github.com/MrWhosNexus/Aether-HE). Thank you - the board is already registered with a drawn layout. But before Aether can actually drive lighting or actuation on your hardware, it needs a real capture of what the official app sends over USB. Nobody can write correct firmware commands by guessing - reusing another board's protocol without proof breaks things, so Aether refuses to mark any board as usable until it has been captured. This guide gets you there in about 10-20 minutes.

Why this step is required

A shared USB vendor ID (VID) means shared OEM silicon, not a shared protocol. Two boards from the same chip vendor can speak completely different command layouts. Aether has a hard rule, locked by an automated test in the codebase: a board is only marked 'drivable' once its own protocol has been captured and decoded. This has already caught a real bug (a board wrongly marked drivable by assuming it shared a sibling board's protocol) - so it is not a formality, it protects your hardware from receiving malformed commands.

Until a capture exists for your board, Aether can draw its layout and remap keys, but cannot safely enable lighting, per-key RGB, or actuation.

Two ways to get this done

- Option A - Manual: follow the steps yourself with the tools below (~15-20 min).
- Option B - Express: hand these same steps to an AI CLI assistant (Claude Code / Fable, Codex/agy, or any Claude/GPT API wrapper such as Hermes) and let it drive the capture with you, then format the result for the GitHub issue.

Either path produces the same two files Aether needs: an HID interface dump, and a USB protocol capture. Pick whichever you're more comfortable with.

Option A: Manual capture

Step 1 - HID interface dump

This tells Aether which USB interface on your board is the 'vendor config' channel that lighting/actuation commands go through.

From the Aether repo root (Python + requirements.txt installed):

```
python tools/list_hid.py YOUR_VID_HEX  
# example: python tools/list_hid.py 0C45
```

Copy the entire output. The interface with usage_page 0xFF00 or higher is almost always the vendor collection - that line matters most, but paste it all.

Step 2 - Protocol capture

This is the important one: a recording of the exact bytes the official app sends to your keyboard when you change a setting.

If your board's software runs inside Chrome or Edge (WebHID):

- Open a new tab to `chrome://device-log`
- Open the official web app and connect your keyboard
- Change one setting - e.g. set a solid color, or change the actuation point
- Copy every 'sendReport' / HID line that appears in the device log around that moment

If your board uses a native desktop app (not a browser):

- Install Wireshark + the USBPcap extension (Windows) - <https://wiki.wireshark.org/CaptureSetup/USB>
- On Linux, use usbmon with Wireshark instead
- Start a capture, filter to your keyboard's USB device, then open the official app and change one setting (color, effect, or actuation point)
- Stop the capture and save the .pcap/.pcapng file

Do this once for lighting, and once for actuation if your board has Hall-effect switches.

Step 3 - Submit it

Attach both the HID dump text and the capture file(s) as a comment on your board's GitHub issue at github.com/MrWhosNexus/Aether-HE. Even a single clean 'set solid red' capture is enough to start decoding.

Option B: The express way (let an AI do the dirty work)

If you don't want to hand-run Wireshark filters or read USB report dumps yourself, you can hand this entire guide to an AI coding assistant running on your own machine and have it walk you through - or drive - the capture. This works with any of the following:

- Claude Code / Fable (Anthropic's CLI agent) - has file, terminal, and browser access
- Codex CLI / agy - OpenAI's equivalent terminal agent
- A lightweight API wrapper you run yourself (e.g. a 'Hermes'-style script calling the Claude or GPT API with terminal tool access)

What to paste into the assistant

Open a terminal in a folder on your computer and start your assistant of choice, then paste this prompt:

```
I'm submitting a keyboard to an open-source project called Aether
(github.com/MrWhosNexus/Aether-HE). I need to produce two things:
  1. An HID interface dump of my keyboard
  2. A USB protocol capture of what the official software sends when
    I change the lighting color (and actuation point, if it's a
    Hall-effect / magnetic switch board).

My keyboard's official software is: <name the app / driver you use>.
My OS is: <Windows / Linux / macOS>.

Please check what tools I have installed (Python, Wireshark, USBPcap,
or usbmon), install anything missing, help me identify my keyboard's
VID/PID, and then walk me step by step through capturing one clean
'set a solid color' USB trace. Explain each step before running it,
since this touches my hardware and installs software.
```

Any competent coding-agent CLI can install Wireshark/USBPcap, find your VID/PID via list_hid.py-equivalent commands, start a capture, and tell you exactly when to click 'apply' in your keyboard's app. It cannot click the button in your keyboard's software for you - that one physical action is still yours.

After the capture

Ask the assistant to summarize the capture file into a short text block (the relevant HID report bytes, not the whole raw pcap), then paste both the summary and the raw capture file into a comment on your board's GitHub issue. If you're not sure how, just ask the assistant: "format this for a GitHub issue comment."

A note on trust

Only run Wireshark/USBPcap installs and captures on a machine and keyboard you own. An AI assistant with terminal access should explain what it's about to install or run before doing it - if it doesn't, ask it to.

You are not sending any of this data anywhere except your own GitHub issue comment; the capture only ever touches your machine until you choose to attach it.

Quick reference

Files Aether needs from you

- HID interface dump (text) - which endpoint to talk to
- One USB capture for lighting (required)
- One USB capture for actuation, if Hall-effect (recommended)

Links

Full submitter doc: docs/SUBMIT_A_BOARD.md in the Aether-HE repo

Board issue tracker: github.com/MrWhosNexus/Aether-HE/issues

Wireshark USB capture setup: wiki.wireshark.org/CaptureSetup/USB

Generated for Aether HE board submitters. Questions? Comment on your board's issue and the maintainer will follow up.